

BOLLAPALLY BANSI DEEPAK

AI Engineer / Full Stack Developer

+91-7330056795 @bansideepak2000@gmail.com Hyderabad, India BansiDeepak Portfolio

PROFESSIONAL SUMMARY

Full Stack Developer and AI Engineer specializing in building end-to-end web applications and integrating AI/ML systems. Proficient in **React**, **Next.js**, **Node.js**, and **TypeScript** for front-end and back-end development, with deep expertise in **LLM frameworks**, **LangChain**, **RAG Architectures**, and deploying production-grade AI platforms using **FastAPI** and **microservices**.

PROFESSIONAL EXPERIENCE

AI Engineer

Neurom Innovations

Dec 2024 – Present Hyderabad, India

- Architected end-to-end RAG systems using schema-aware retrieval, enabling high-accuracy LLM responses for enterprise data sources.
- Built a dynamic multi-model inference gateway in **FastAPI** that intelligently routes traffic between **Google Gemini** and **OpenAI** based on task complexity, latency budget, and cost-efficiency.
- Reduced hallucinations by designing enterprise-grade guardrails, including policy-driven validation, tool-calling restrictions, and structured output enforcement.
- Led full-stack development of internal AI pilots, including React/Next.js dashboards, asynchronous inference queues, and real-time task monitoring.
- Delivered full-stack web applications across multiple client projects — building React/Next.js frontends, FastAPI/Express backends, and integrating AI systems end-to-end.
- Optimized LLM performance pipelines, benchmarking inference cost, throughput, caching layers, and implementing structured prompting frameworks.

TECHNICAL SKILLS

React.js

Next.js

TypeScript

JavaScript

Node.js

Vite

Tailwind CSS

Streamlit

FastAPI

Python

Google Gemini

OpenAI API

LangChain

RAG Pipelines

Vector Search

Prompt Engineering

Text-to-SQL

Plotly

Pandas

NumPy

PostgreSQL

Docker

Kubernetes

AWS Amplify

AWS EC2

Git

CI/CD

KEY PROJECTS

ASKENCE – Enterprise Text-to-SQL & Analytics Platform

React 19, TypeScript, FastAPI, LangChain, Gemini, PostgreSQL, Docker

- Designed an advanced Text-to-SQL RAG system using **LangChain** and **Schema Injection** to generate accurate and executable SQL queries from natural language.
- Built a **SQL Validation & Rewriting Engine** that detects large datasets and automatically produces optimized, aggregated queries.
- Developed a dynamic visualization engine using **React + TypeScript + Vite** that classifies output types and renders appropriate visualizations.
- Deployed system for **real-world enterprise** data analytics, enabling business users to query operational data through natural language.

SEARCHLIFT360 – Asynchronous AI SEO Auditing Engine

FastAPI, Python, Docker, Next.js 14, AWS Amplify

- Architected a scalable asynchronous crawling engine using **FastAPI Background Tasks**, enabling parallel deep-site audits with zero UI blocking.
- Developed a unique hybrid analysis pipeline combining **rule-based DOM Checks** with **generative AI insights** for content strategy.
- Automated the creation of client-ready PDF audit reports using **ReportLab**, generating structured, visual SEO health summaries.

AI Call Automation

React, Python, OpenAI, Twilio, AWS Bedrock

- Developed a **React**-based tool for comparing **OpenAI**, **Vertex AI**, and **Bedrock** models with consistent system prompts.
- Implemented **Twilio** integration with Speech-to-Text (STT) and Text-to-Speech (TTS) for automated conversational flows.
- Deployed AI models using **AWS Bedrock** and managed compute resources via **EC2**, ensuring scalable performance.

Neurom.com – Company Landing Page

Next.js 15, Sanity CMS, GSAP, Framer Motion, Tailwind CSS v4

- Built a company landing page with headless CMS-powered content management using **Next.js 15** and **Sanity CMS**.
- Implemented rich animations with **GSAP** and **Framer Motion**, styled with **Tailwind CSS v4**.

CERTIFICATIONS

- Oracle Cloud Infrastructure 2023 Certified Associate
- Postman API Fundamentals Student Expert

EDUCATION

B.Tech, Computer Science & Engineering

Talla Padmavathi College of Engineering

2021 – 2025